

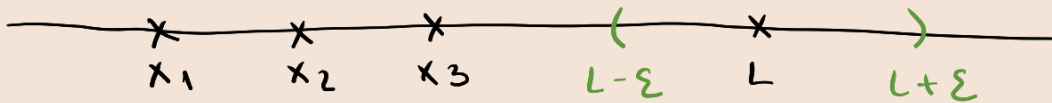
Seminar 2 (14.10.2025)

① Prove using the ε definition that $\lim_n \frac{1}{\sqrt{n}} = 0$.

($\forall$) $\varepsilon > 0$, $\exists N_\varepsilon \in \mathbb{N}$ s.t. $|x_m - l| < \varepsilon$, ($\forall$) $m \geq N_\varepsilon$.

$$\frac{1}{\sqrt{m}} = x_m; \lim_n x_m = \lim_n \frac{1}{\sqrt{m}} = 0 = l.$$

$$x_1, x_2, \dots, x_m, \quad x_i \in \mathbb{R}.$$



($\forall$) $\varepsilon > 0$, $\exists N_\varepsilon \in \mathbb{N}$ s.t. $\left| \frac{1}{\sqrt{m}} - 0 \right| < \varepsilon$, ($\forall$) $m \geq N_\varepsilon$

$$\Leftrightarrow \underbrace{\left| \frac{1}{\sqrt{m}} \right|}_{>0} < \varepsilon, \quad (\forall) m \geq N_\varepsilon$$

$$\Rightarrow \frac{1}{\sqrt{m}} < \varepsilon \Rightarrow \frac{1}{\varepsilon} < \sqrt{m} \quad \Big|^2 \Rightarrow \boxed{\frac{1}{\varepsilon^2} < m}$$

$$N_\varepsilon = \left\lceil \frac{1}{\varepsilon^2} \right\rceil + 1 \Rightarrow \frac{1}{\varepsilon^2} < m, \quad (\forall) m \geq N_\varepsilon.$$

$$(\varepsilon = 2 \Rightarrow \frac{1}{\varepsilon^2} = \frac{1}{4} = 0,25; \left\lceil \frac{1}{\varepsilon^2} \right\rceil = 0 + 1 = 1)$$

② Study if the sequence is bounded, monotone and convergent.

$$a) \quad \sqrt{n+1} - \sqrt{n} = x_n$$

$$x_1 = \sqrt{2} - \sqrt{1} \approx 0,4$$

$$x_2 = \sqrt{3} - \sqrt{2} \approx 0,3$$

$\vdots$

$$x_m = \sqrt{m+1} - \sqrt{m}$$

$$x_{m+1} \geq x_m \quad \swarrow \quad \searrow \quad x_{m+1} \leq x_m$$

$$0 \leq x_m \Rightarrow 0 \text{ is lb for } (x_m) \quad (1)$$

$$\begin{aligned} x_{m+1} - x_m &= \sqrt{m+2} - \sqrt{m+1} - \sqrt{m+1} + \sqrt{m} \\ &= \sqrt{m+2} - 2\sqrt{m+1} + \sqrt{m} \stackrel{?}{<} 0 \end{aligned}$$

$$\Rightarrow \sqrt{n+2} + \sqrt{n} < 2\sqrt{n+1} \quad |^2$$

$$\Rightarrow 2n+2 + 2\sqrt{n(n+2)} < 4(n+1) \quad | -2n$$

$$\Rightarrow 2\sqrt{n(n+2)} < 2n+2 \quad | :2$$

$$\Rightarrow \sqrt{n(n+2)} < n+1 \quad |^2$$

$$\Rightarrow n^2 + 2n < n^2 + 2n + 1, (\forall) n \in \mathbb{N}$$

$$\Rightarrow x_{n+1} - x_n < 0 \Rightarrow x_{n+1} < x_n$$

$$\Rightarrow (x_n) \text{ decreasing (2)}$$

(1), (2) $\Rightarrow (x_n)$ convergent.

$$\lim_n (\sqrt{n+1} - \sqrt{n}) = \lim_n \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \lim_n \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0 \in \mathbb{R}.$$

The limit $\exists$, $\Rightarrow$ convergent.
finite

③ Find the limit

$$c) \lim_n \sqrt[n]{n} = \lim_n e^{\frac{1}{n} \ln n} = e^{\lim_n \frac{\ln n}{n}} \stackrel{(*)}{=} e^0 = 1.$$

$$(*): \lim_n \frac{\ln n}{n} = 0$$

$$\ln / e^n > n, (\forall) n > 1$$

$$\Rightarrow n > \ln n \leadsto \sqrt{n} > \ln \sqrt{n}$$

$$0 \leq \frac{\ln n}{n} \leq \frac{2 \ln \sqrt{n}}{n} \leq \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}}$$

$$\searrow \quad \searrow \quad \searrow$$

why does $\ln n$ go slower to ∞ than n

$$e) \lim_n (a_1^n + a_2^n + \dots + a_k^n)^{\frac{1}{n}}, a_i > 0.$$

$$\text{Ex: } \lim_n (2^n + 3^n)^{\frac{1}{n}} = \lim_n \left(3^n \left(\frac{2^n}{3^n} + 1 \right) \right)^{\frac{1}{n}} =$$

$$= \lim_n 3 \cdot \left(\underbrace{\left(\frac{2}{3} \right)^n}_{\rightarrow 0} + 1 \right)^{\frac{1}{n}} = 3 \cdot (0+1)^0 = 3.$$

Assume that $a_1 \geq a_2 \geq \dots \geq a_k$.

$$\begin{aligned} x_n &= (a_1^n + a_2^n + \dots + a_k^n)^{\frac{1}{n}} \\ &= \left[a_1^n \cdot \left(1 + \underbrace{\left(\frac{a_2}{a_1} \right)^n}_{\substack{\leq 1 \\ \geq 0}} + \underbrace{\left(\frac{a_3}{a_1} \right)^n}_{\substack{\leq 1 \\ \geq 0}} + \dots + \underbrace{\left(\frac{a_k}{a_1} \right)^n}_{\substack{\leq 1 \\ \geq 0}} \right) \right]^{\frac{1}{n}} \end{aligned}$$

$$\lim_n a_1 (1 + \underbrace{0 + \dots + 0}_{k-1})^{\frac{1}{n}} \leq \lim_n x_n \leq \lim_n a_1 (1 + \underbrace{1 + 1 + \dots + 1}_{k-1})^{\frac{1}{n}}$$

$$\Leftrightarrow \lim_n a_1 \leq \lim_n x_n \leq \lim_n a_1 \cdot \underbrace{k^{\frac{1}{n}}}_{\rightarrow 1}$$

$\swarrow \quad \downarrow \quad \searrow$
 $\rightarrow a_1$

$$\Rightarrow \lim_n x_n = a_1 = \max \{a_1, \dots, a_k\}$$

④ $e_n = \left(1 + \frac{1}{n}\right)^n$. Prove that (e_n) is increasing and bounded, hence convergent.

• e_n is bounded from above
 (from below is obvious $\rightarrow 0$)

$$\boxed{(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k}$$

$\downarrow$
 $\frac{n!}{(n-k)! \cdot k!}$

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \cdot 1 \cdot \frac{1}{n^k}$$

$$= 1 + n \cdot \frac{1}{n} + \binom{n}{2} \cdot \frac{1}{n^2} + \dots + \binom{n}{n} \cdot \frac{1}{n^n}$$

$$= 1 + 1 + \frac{n(n-1)}{2!} \cdot \frac{1}{n^2} + \dots + \frac{n(n-1) \dots (n-k+1)}{k! \cdot n^k} + \dots + \frac{1}{n^n}$$

$$\begin{aligned}
&= 1 + 1 + \frac{1}{2} \left(\frac{m^2 - m}{m^2} \right) + \dots + \frac{1}{k!} \left(\frac{m(m-1) \dots (m-k+1)}{m \cdot m \dots m} \right) + \dots + \frac{1}{n^3} \\
&= 1 + 1 + \frac{1}{2} \left(1 - \frac{1}{m} \right) + \dots + \frac{1}{k!} \cdot \frac{m}{m} \cdot \frac{m-1}{m} \dots \frac{m-k+1}{m} + \dots + \frac{1}{n^3} \\
&= 1 + 1 + \frac{1}{2!} \underbrace{\left(1 - \frac{1}{m} \right)}_{< 1} + \dots + \frac{1}{k!} \underbrace{\left(1 - \frac{1}{m} \right)}_{< 1} \underbrace{\left(1 - \frac{2}{m} \right)}_{< 1} \dots \underbrace{\left(1 - \frac{k-1}{m} \right)}_{< 1} + \dots + \frac{1}{n^3}
\end{aligned}$$

! multiplying by sth < 1 makes it smaller

$$\left. \begin{array}{l} m^m = m \cdot m \dots m \\ m! = 1 \cdot 2 \dots m \end{array} \right\} \Rightarrow m^m > m! \Rightarrow \frac{1}{m^m} < \frac{1}{m!}$$

$$1 + 1 + \frac{1}{2!} \underbrace{\left(1 - \frac{1}{m} \right)}_{< 1} + \dots + \frac{1}{k!} \underbrace{\left(1 - \frac{1}{m} \right)}_{< 1} \underbrace{\left(1 - \frac{2}{m} \right)}_{< 1} \dots \underbrace{\left(1 - \frac{k-1}{m} \right)}_{< 1} + \dots + \frac{1}{n^3}$$

$$1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \dots + \frac{1}{n!}$$

$\left. \begin{array}{l} k! = 1 \cdot 2 \dots k \\ 2^{k-1} = \underbrace{1 \cdot 2 \cdot 2 \dots 2}_k \end{array} \right\} \Rightarrow k! > 2^{k-1} \Rightarrow \frac{1}{k!} < \frac{1}{2^{k-1}}$

$$1 + 1 + \frac{1}{2^{2-1}} + \frac{1}{2^{3-1}} + \dots + \frac{1}{2^{k-1}} + \dots + \frac{1}{2^{n-1}}$$

$2 + \frac{1}{2} = 3 \Rightarrow (e_n)$ is bounded (1)

$$\frac{1}{2^1} + \frac{1}{2^2} + \dots + \frac{1}{2^{k-1}} + \dots + \frac{1}{2^{n-1}} < 1 \quad ?$$

• (e_n) is increasing: $e_{n+1} \stackrel{?}{\geq} e_n$
 $\Rightarrow \left(1 + \frac{1}{n+1} \right)^{n+1} \stackrel{?}{\geq} \left(1 + \frac{1}{n} \right)^n$

$$\boxed{\sqrt{a \cdot b} \stackrel{IH}{\leq} \frac{a+b}{2}}$$

$$\boxed{\sqrt[n+1]{x_1 \cdot x_2 \dots x_n \cdot x_{n+1}} \stackrel{IH}{\leq} \frac{x_1 + x_2 + \dots + x_n + x_{n+1}}{n+1}}$$

Let $x_1 = 1$,

$$x_2 = x_3 = \dots = x_{n+1} = 1 + \frac{1}{n}$$

$\downarrow$
 n times

$$\sqrt[n+1]{1 \underbrace{\left(1 + \frac{1}{n}\right) \cdot \dots \cdot \left(1 + \frac{1}{n}\right)}_{n \text{ times}}} \stackrel{IH}{\leq} \frac{1 + \underbrace{\left(1 + \frac{1}{n}\right) + \dots + \left(1 + \frac{1}{n}\right)}_{n \text{ times}}}{n+1}$$

$$\Rightarrow \sqrt[n+1]{1 \cdot \left(1 + \frac{1}{n}\right)^n} \stackrel{IH}{\leq} \frac{1 + n \cdot \left(1 + \frac{1}{n}\right)}{n+1} \quad |^{n+1}$$

$$\Rightarrow \left(1 + \frac{1}{n}\right)^{\frac{n}{n+1}} \stackrel{IH}{\leq} \left(\frac{1 + 1 + n}{n+1}\right)^{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1}$$

$$\Rightarrow e_n \leq e_{n+1} \Rightarrow (e_n) \text{ is increasing (2)}$$

(1), (2) $\Rightarrow (e_n)$ convergent.